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The Grammar of Perception: A Case Study of Guambiano (Namtrik) and Duna

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Abstract

Although the senses provide a common basis for our experience of the world, cross-linguistic research has revealed substantial diversity across languages in how perceptual experiences are encoded in words. Such variability is also present at the level of grammatical structure, though the project of charting the parameters and limits of cross-linguistic diversity at this level is still in its relative infancy. We contribute to this goal by presenting a comparative study of the grammar of perception in two understudied and endangered languages, Guambiano (a Barbacoan language spoken in southern Colombia), and Duna (a Duna-Bogaia language of the Papua New Guinean highlands). Our study is based on primary fieldwork and draws on spoken language data collected using the *Grammar of Perception Questionnaire*. We document the lexical and constructional resources available in the two languages for describing events of perception and examine how sensory meanings are packaged and distributed across different elements of these constructions. The results reveal previously undocumented patterns

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of perception event encoding, underscoring the multifaceted and complex relationship between language and perception.

Keywords: Perception; Linguistic diversity; Grammar; Sensory linguistics; Evidentials; Subjectivity

1. Introduction

The cross-linguistic study of perceptual language gives immense scope for examining how language interfaces with cognition (e.g., Evans & Wilkins, 2000; Levinson & Majid, 2014; Malt & Majid, 2013), and research in this area has underscored the imperative to integrate linguistic diversity into cognitive science (Majid & Levinson, 2011; Majid et al., 2018). Elements of language that encode perceptual meanings must map onto the world as interpreted through our perceptual organs, subserved by a shared biology and cognitive architecture. However, time and again cross-linguistic research has shown that perceptual language is nowhere near as uniform as this might predict and may in turn even exert its own influences on non-linguistic cognition (Lupyan, Rahman, Boroditsky, & Clark, 2020).

Work with speakers of lesser-studied languages, using a variety of methods, has on the one hand supported the idea that common properties of sensory cognition constrain linguistic categories, for example, as observable in fascinating parallelisms between the lexicon and perceptual psychophysics. In the language Maniq, olfaction verbs may be partly structured around properties such as evaluation and edibility, which can be significant factors in odor perception and categorization (Wnuk & Majid, 2012); and in Siwu, the rich and semantically complex word class of ideophones nevertheless cleaves in part to psychophysically salient haptic dimensions, such as roughness and springiness (Dingemanse & Majid, 2012). On the other hand, case studies have also demonstrated remarkable diversity in the ways in which languages carve up the sensorium. Umpila (Cape York, Australia), for example, has only three basic color words, while English has nearly a dozen (Hill, 2011). And while English must make do with only a few dedicated terms to describe olfactory properties, others have rich and sizeable smell lexicons (Burenhult & Majid, 2011; Majid & Burenhult, 2014; O'Meara & Majid, 2016; Wnuk & Majid, 2014; Wnuk, Laophaioj, & Majid, 2020). Such differences, which appear to be shaped by diverse cultural and environmental factors (Floyd, San Roque & Majid, 2018; Majid, 2015; Majid & Kruspe, 2018; Majid et al., 2018) have wide-ranging consequences for understanding diversity and plasticity in cognition, for example, in relation to the potential for linguistic encoding to influence memory and perceptual discrimination (e.g., Lupyan et al., 2020; Roberson, Pak, & Hanley, 2008; Thierry, Athanasopoulos, Wiggett, Dering, & Kuipers, 2009).

To date, cross-linguistic research on regularity and diversity in perceptual language has largely centered on the lexicon (see Majid & Norcliffe, 2026, for an overview). For some perceptual domains, this is an inevitable result of the linguistic affordances of the domain itself—semantic distinctions for color, temperature, and texture, for example, appear to be entirely restricted to the lexical level (Allan, 1977; Evans, 2011; Goldin-Meadow, 2007). Other domains show more fluidity. This is found, for example, for certain sensory property

domains, such as shape (Friedrich, 1970) and (rarely) smell (Floyd, San Roque & Majid, 2018), but is perhaps most salient for the linguistic encoding of perception *events*. At the lexical level, perception events are labeled by perception verbs, such as *see*, *hear*, *smell*, which describe how people experience the world through their senses. All languages appear to have at least one perception verb, and these have been the subject of considerable cross-linguistic research since Viberg's (1983) foundational typology. This work has highlighted both systematicity and variability across languages in the elaboration and structuring of perception verb inventories (Evans & Wilkins, 2000; Georgakopoulos, Grossman, Nikolaev, & Polis, 2022; Norcliffe & Majid, 2024a; Norcliffe & Majid, 2024b; Sweetser, 1990; Vanhove, 2008). For example, vision is the sense most likely to have a dedicated verb in the lexicon (Evans & Wilkins 2000; Norcliffe & Majid, 2024a; Viberg, 1983), and visual verbs and vocabulary are more frequent in speech and learned earlier by children, compared to the other senses (e.g., San Roque et al., 2015; San Roque, Norcliffe, & Majid, 2024; Viberg, 1994; Winter, Perlman, & Majid, 2018), plausibly reflecting the dominance of vision in human perception. In other ways, verbs are a site of variation, for example, in the specific patterns of polysemy that are encoded across languages (Evans & Wilkins, 2000), underscoring the powerful role of culture in shaping perceptual vocabularies. However, thinking of verbs as isolated single words does not capture their role within the complex grammatical ecology of perception event descriptions, for example, the choice between presenting a perceiver or a stimulus as the focal element (i.e., clausal subject) of a sensory phenomenon. And, working in tandem with basic perception verbs, languages potentially draw on a range of grammatical resources to encode aspects of perception. As a parallel within the related domain of emotion, for example, key meanings (e.g. surprise) and contrasts (e.g. physical vs. emotional states) may be expressed by morphological and syntactic, rather than lexical, means (Majid, 2012a). However, we still know relatively little about how languages carve up perceptual experience in grammar or about the division of labor between lexical and grammatical levels.

As a small contribution to this bigger enquiry, in this paper, we examine the lexical and grammatical expression of perception through two understudied and unrelated languages: Guambiano (Namtrik), a Barbacoan language spoken in southern Colombia, and Duna, a Duna-Bogaia language of the Papua New Guinean highlands. We bring these languages together in large part due to our own respective expertise, though they serendipitously share certain lexical characteristics (e.g., relatively small inventories of basic perception verbs) as well as grammatical features that are not commonly found in better studied European languages and yet are highly relevant to the expression of perception (e.g., evidential marking, which can require speakers to signal their source of knowledge). Considered together, the languages afford the opportunity to closely examine how features of linguistic systems that are rather different to English, but relatively similar to each other, may be co-opted into describing events of perception. In the exploratory process of charting two particular instantiations of perception grammar, we aim to identify, where possible, lines of pursuit for future targeted research, for example, concerning the grammatical expression of multi-modal sensory experience, the spatial encoding of perception, or the functional (non-)equivalence of lexical and grammatical elements in perception descriptions (Majid, 2012a; Mushin, 2013; Squartini, 2018). Our case study illustrates that the comparison of perceptual language shared with us by

two speakers from just two of the world's 7000 languages both contributes new language facts and sparks new questions about interactions between sensory experience, conceptualization, and language structure and that this is part of a rapidly dwindling embarrassment of riches, given that more than half of those languages are estimated to be endangered (Bromham et al., 2021).

Our language consultants contributed information, examples, and expertise through undertaking the *Grammar of Perception Questionnaire* (GPQ; Norcliffe, Enfield, Majid, & Levinson, 2010), a translation questionnaire covering a range of perception event types according to a standard typological framework (Viberg, 1983, Section 3). Using a translation questionnaire supported comparison of grammatical strategies used to describe the same simple perception events and provided coverage across different sensory modalities, including smell and taste, that are less frequent in natural conversation in many languages (San Roque et al., 2015). The questionnaire builds on Viberg's (1983) approach to the typology of perception events, with a focus on the semantic elements of CONTROL, SENSE MODALITY, and ORIENTATION. Control relates to the ambiguous status of sensory experience as something that "comes upon us" as opposed to an investigative power that we consciously direct (as, for instance, lexicalized in the English distinction between *see* and *look*), and thus relates to the expression of agency and attention. Sense modality concerns how (and if) language distinguishes between different sensory channels, separating continuous and multimodal embodied sensation into distinct lexical components (e.g., *hear* vs. *feel*). Orientation refers to whether a perception event is encoded as reception, with a focus on the animate, sensing perceiver as the subject of the clause (e.g., *the cat saw the king*), or with a focus on the percept as a subject that emanates or "gives off" qualia to be experienced (e.g., *the king looked nervous*).

We provide background information on Guambiano and Duna (Section 2), and then introduce Viberg's typological framework and the GPQ (Section 3). In Section 4, we sketch the basic perception verbs of our two focus languages, noting how their lexicons reflect or refract Viberg's framework. We then move beyond the lexicon to consider certain ways in which Guambiano and Duna deploy grammaticalized or constructional resources to express control, modality, and orientation (Section 5). In Section 6, we briefly consider an aspect of perception events not featured in the Vibergian framework, but which can play an important role in grammar: the inherent subjectivity of perception, and grammatical sensitivity as to who can assert perceptual experiences with authority. Section 7 summarizes our findings and offers suggestions for further study. Overall, our study of Guambiano and Duna perception grammar emphasizes the multidimensionality of expressing perceptual experience and the variety of grammatical resources that we can draw on to accomplish this task, revealing a small fragment of the immense variation that motivates and challenges cognitive science.

2. Language background

Guambiano belongs to the Barbacoan language family, a small group of languages spoken in southern Colombia and the coastal provinces of Ecuador. Guambiano, whose speakers number around 21,000 (Simons & Fennig, 2017), is also known endogenously as Namtrik

(‘our mouth’) or Namuy Wam (‘our sound’); the term Namtrik is also sometimes used as an umbrella term for the different varieties of the language (Gonzales Castaño, 2019), spoken in the *resguardos* (state-recognized territorial entities) of Guambía, Quisgó, Ambaló, and Totoró, within the Colombian department of Cauca. The extent of the differences between language varieties is not yet established (Gonzales Castaño, 2012, p. 215).¹ Some characteristic grammatical features of Namtrik/Guambiano include Subject Object Verb (SOV) basic word order, accusative case, switch reference, and evidential marking. Bilingualism with Spanish is widespread, with rates varying by generation as well as community (Gonzales Castaño, 2019). The data for the present study were collected by the first author with a Guambiano (Guambía dialect) speaking member of the Misak community of Fontibón, Bogotá. This small, close-knit community of around 200 inhabitants has formed within the last 20 or so years as a response to increasing scarcity of farmable land and continuing narcotrafficking-related violence in their traditional homelands. The perception verb questionnaire and the accompanying meta-discussion were conducted orally in Spanish, in an informal setting. Responses were collected for the basic questionnaire scenarios as well as some supplementary sections. In various places, the questionnaire prompted further discussion of aspects of Guambiano grammar, which elicited further relevant examples. We also draw on these, where relevant to the discussion. The Guambiano analyses are informed by earlier fieldwork as well as previously published research on the language (Floyd & Norcliffe, 2016; Norcliffe, 2018; Vásquez de Ruíz, 1988; Vásquez de Ruíz, 2007.)

Duna (Yuna) is a Trans New Guinea (TNG) language (Evans & Fedden, in press) spoken by some tens of thousands of people local to the Lake Kopyago region in Papua New Guinea. Most Duna speakers combine subsistence gardening with small-scale and sporadic economic enterprise and face complex challenges regarding equitable access to health, education, economic and personal security, and political representation. Duna shares many typical characteristics of TNG languages, such as verb-final sentence structure, extensive use of verb serialization and clause chaining, and pragmatically moderated “ergative” marking. More unusual elements of the language within the TNG context include a complex evidential system (an areal feature, Foley, 1986; San Roque & Loughnane, 2012). The language is under-described, under-documented, and endangered. Most Duna speakers are multilingual with at least one other Indigenous language and/or the lingua franca Tok Pisin (an English-based creole) in their repertoire, as well as potentially some English. The data discussed in this paper were collected by the second author with a Duna speaker from Rale (PK), who had extensive experience in language consultation and translation work, and the collection setting was informal and discursive. Questionnaire prompts (delivered orally but with the written English text also available to PK) and accompanying meta-discussion took place largely in Tok Pisin, with some use of English and Duna. As with Guambiano, responses were collected for the basic scenarios and some supplementary sections (e.g., “want to know” frames). For some scenarios, PK also made judgments or commented on alternative translations offered by the researcher (discussed as relevant in the following sections). The Duna analyses are also informed by earlier fieldwork and discussion with a range of other Duna speakers from the region (e.g., San Roque, 2008).

Table 1
“Viberg Table” for English verbs of perception

	Perceiver-Oriented		Stimulus-Oriented
	Controlled	Non-Controlled	
Visual	<i>look</i>	<i>see</i>	<i>look</i>
Auditory	<i>listen</i>	<i>hear</i>	<i>sound</i>
Tactile	<i>feel, touch</i>	<i>feel</i>	<i>feel</i>
Gustatory	<i>taste</i>	<i>taste</i>	<i>taste</i>
Olfactory	<i>smell</i>	<i>smell</i>	<i>smell</i>

3. Viberg’s framework and the GPQ

The GPQ is designed to gather cross-linguistically comparable data on grammatical strategies that are used to describe perceptual events and states (Norcliffe et al., 2010). It builds on previous field tasks in the domain of sensory language published by the Language and Cognition Department at the Max Planck Institute for Psycholinguistics.

The GPQ is structured to align with key semantic components of perception events as conceptualized in Viberg’s (1983, 2001) typological framework developed for his study of perception verbs in 53 languages. Across languages, there is considerable variability in how perception verbs carve up this semantic space (Norcliffe & Majid, 2024a; Viberg, 1983). Viberg identified three components cross-linguistically relevant to perception verb lexicalization: control, orientation, and sense modality, presenting this as a workable “standard analysis” (p. 124) that drew on prior work such as Buck (1949). Table 1 illustrates how these semantic components are lexically distinguished (or collapsed) in basic perception verbs in English, discussed further below.

Control refers to the agentivity and intentionality of the perceiver. A perceiver may notice sensory data but do so accidentally or, alternatively, may direct purposeful attention via a sensory channel in order to perceive. English has distinct “controlled” (also termed “activity”) verbs in some sense modalities (e.g., *look, listen, touch*), contrasting with “non-controlled” or “experience” verbs (e.g., *see, hear*). Orientation refers to the way that the participant roles of perceiver and stimulus are mapped onto the grammatical role of subject. Whereas controlled and non-controlled verbs orient to the perceiver as subject (and are typically transitive, taking the stimulus as an object), stimulus-based verbs are typically intransitive and orient to the stimulus as the subject (e.g., *his voice sounds scratchy*). This latter class appears to be only inconsistently lexicalized within perception verbs across languages (Aikhenvald & Storch, 2013; Norcliffe & Majid, 2024a; see also Viberg, 2019).

Cross-cutting these domain-general distinctions is the category of sense modality, which distinguishes between visual, auditory, tactile, gustatory, and olfactory modalities. Scientifically, this division is a simplification (see O’Meara, Speed, San Roque, & Majid, 2019; Winter, 2019, for discussion), reflecting a “common sense” categorization (folk model) of perceptual experiences based on their association with a dedicated sensory organ (seeing involves the

eyes, hearing involves the ears, etc.). It is not the only model, as sensory anthropologists have long pointed out (Classen, 1997), but many unrelated languages do distinguish between the five sense modalities in their perception verb lexicons (Norcliffe & Majid, 2024a), indicating that such a conceptualization of the senses is widespread.

The GPQ incorporates these semantic components in a set of 60 main scenarios and 63 supplementary scenarios. Each scenario consists of an initial context (to be adapted in ways appropriate to the language community), followed by a target clause or sentence to be translated. An example is (E1), with the translation target highlighted in bold:

(E1)² My mother had baked some bread. She gave me some. **It tasted delicious.**

For each event type (controlled, non-controlled, etc.), the GPQ systematically includes translation scenarios for each sensory modality. In this way, it can address the question of whether lexical and grammatical resources (or constraints) apply differently according to sensory modality, and how (if at all) languages might convey such sensory meanings in cases where they are not lexically distinguished in verbs. The questionnaire does not cover other kinds of sensory distinctions that are known to be lexicalized in some languages, such as proprioception and interoception; this remains an important area for future comparative study.

The questionnaire also includes contexts that capture other features of perception events such as evaluative content (e.g., expressing that a smell is pleasant or unpleasant) and person asymmetries (whether, e.g., first person descriptions of perception events systematically differ from non-first person descriptions; see, e.g., Floyd, Norcliffe & San Roque, 2018).

Translation questionnaires have limitations (see, e.g., Majid, 2012b). Speakers may be influenced by the grammatical structure of the source language, for example, and they lack the contextual richness of real perception events and the ecological validity of spontaneous language. Nevertheless, they are useful for obtaining an initial comparative picture, especially in the case of low-resource languages where large corpora are lacking. We leave it to future work to supplement and refine the observations presented here with methods that can capture contextually and culturally embedded use of perception-related expressions.

4. Verbs of perception in Guambiano and Duna

We begin by examining how Guambiano and Duna lexicalize the semantic components of control, sense modality, and orientation in perception verbs. Guambiano and Duna's set of basic perception verbs are presented in the "Viberg" tables in Tables 2 and 3. As can be seen, both languages have fewer simple verbs than English.

Guambiano lexically distinguishes between visual (*aship* 'to see, to look') and non-visual perception (*mirip* 'to sense') for perceiver-oriented verbs. The verb *mirip* can express perceptual experiences via any of the non-visual modalities, though depending on the nature of the stimulus, the perception verb may require extra constructional support (see Section 5.1):

Table 2
“Viberg Table” for Guambiano verbs of perception

	Perceiver-Oriented		Stimulus-Oriented
	Controlled	Non-Controlled	
Visual	<i>aship</i>	<i>aship</i>	
Auditory	<i>mirip</i>	<i>mirip</i>	<i>wantrip</i>
Tactile	<i>kemiri mirip</i> ‘touch’ + ‘sense’	<i>mirip</i>	
Gustatory	<i>ma mirip</i> ‘eat’ + ‘sense’ <i>mucha mirip</i> ‘drink’ + ‘sense’	<i>mirip</i>	
Olfactory	<i>itri mirip</i> smell + sense	<i>mirip</i>	<i>itrip</i>

Table 3
“Viberg Table” for Duna verbs of perception

	Perceiver-Oriented		Stimulus-Oriented
	Controlled	Non-Controlled	
Visual	<i>ke-</i>	<i>ke-</i>	
Auditory	<i>waki-</i> <i>waki ke-</i>	<i>waki-</i>	
Tactile	<i>suwa ke-</i> ‘hold see’ <i>ala ka-</i> ‘touch’	<i>waki-</i>	
Gustatory	<i>neya ke-</i> ‘eat see’	(<i>waki-</i>)	
Olfactory	<i>kori ke-/waki</i> <i>ringa ke-/waki</i>	<i>kori-</i> <i>ringa-</i> <i>waki-</i>	<i>kori-</i> ‘smell(.good)’ <i>ringa-</i> ‘smell.bad’

- (G2) *traseik-wan mir-ar*
scream-ACC sense-EGO
‘I heard a scream.’

[auditory]
- (G3) *na isik ichipik-wan mir-ar*
1SG wind blow-ACC sense-EGO
‘I felt the wind blow.’

[tactile]
- (G4) *na nakkisr itrip-en mir-ar*
1SG smoke smell-NOM sense-EGO
‘I smelled smoke.’

[olfactory]

- (G5) *nei mamik-u kirapik wepatrup-en mir-ar*
 1POSS soup-LOC spicy appear-NOM sense-EGO
 ‘I sensed something spicy in my soup.’ [gustatory]
 [Lit. ‘I sensed when something spicy appeared in my soup’]

For vision and audition, there is no lexicalized distinction between controlled versus non-controlled perception:

- (G6) *na kan piyau ash-ar*
 1SG one fish see-EGO [non-controlled vision]
 ‘I saw a fish’
- (G7) *na kusru-a ash-ar*
 1SG lift-SS see-EGO [controlled vision]
 ‘I picked it up and looked at it’

For the modalities of touch, taste, and smell, grammaticalized serial verb constructions are used to express controlled perceptual meanings (printed in gray); we discuss these strategies in Section 5.2. Stimulus-oriented verbs show yet a different dissection of the sensory space: Only the auditory and olfactory modalities have dedicated verbs:

- (G8) *kasrak-kirishik-pe tapchik wantr-an*
 happiness-music-TOP beautiful sound-NON.EGO
 ‘The music sounded beautiful’ [stimulus-oriented, auditory]
- (G9) *mamik maichik itr-an*
 soup delicious smell-NON.EGO
 ‘The soup smelled delicious’ [stimulus-oriented, olfactory]

Similar to Guambiano, non-controlled perception verbs in Duna (Table 3) distinguish between visual (*ke-*) and non-visual perception (*waki-*). From typical usage of *waki-*, as well as people’s informal explanations of its meaning, this verb primarily denotes auditory perception. However, it can also be used to describe tactile and olfactory perception, and potentially taste (D11), although this appears to be very uncommon.

- (D10) *akura kone, no ndune ndu waki-u.*
 night true 1SG thing one hear-PFV
 ‘In the night, I heard something.’ [auditory]
- (D11) *na=yeritia-ko waki-u*
 eat=SNS.PFV-SUB sense-PFV
 ‘I sensed that I ate [chilli].’ [gustatory]

Waki- can also be used to talk about internal feeling, as in English ‘feeling unwell’.

- (D12) *nane Joni=yanua kho puru-ta siki sa=yatia-ko waki-u.*
 boy PSN=SNS.IMPL 3SG body-OBL sickness strike=SNS.IPV-SUB sense-PFV
 ‘This boy John he sensed sickness strike his body.’ [internal feeling]

As in Guambiano, controlled perception for touch, taste, and smell is largely expressed through more complex constructions (verb adjunct or coverb constructions, printed in gray). However, the complex predicate *ala ka-* ‘touch, feel’ (formed with the coverb *ala* and existential verb *ka-* ‘be, stand, make’) implies control or intention as part of its meaning, and this expression was identified by language consultant PK as suitable for purposeful actions, such as (D13), only.

- (D13) *no antia=ka no plenkiya wi hini mo ndu ngu.*
 1SG mother=ERG 1SG blanket possum fur like one give.PFV
na ala ka-yeroko, keko kone=yatia
 1SG.ERG touch be/stand-SW.SIM soft truly=SNS.IPFV
 ‘My mother gave me a blanket like possum fur. I was touching [it], it was very soft.’

Duna does not appear to have a stimulus-oriented verb with a general auditory meaning, although it has a wide range of more specific auditory terms (e.g., ‘whine’, ‘creak’, and ‘howl’) and appearance words (e.g., referring to coloration, patterning, and light qualities). Smell has two dedicated verb roots that also encode an evaluative distinction (good vs. bad-smelling). These verb roots can be used transitively (D14, D16) but (perhaps more typically) as intransitive stimulus-oriented terms (see D15, D17), an option that is lacking for the other senses. While both terms are usually interpreted as evaluative, it seems that, at a pinch, *kori-* can be used as a more neutral smell verb (e.g., see D20), and this requires further study.

- (D14) *Na sopo=na kori-u*
 1SG.ERG soap=SPEC smell.good-PFV
 ‘I smell the (pleasant) soap.’

- (D15) *Ha=na kori-pa*
 HA=SPEC smell.good-IPFV
 ‘That is smelling nice.’

(D16) *na wena ndu ringa-o*
 1 SG.ERG fish one smell.bad-PFV
 ‘I smell a (stinky) fish.’

(D17) *Hinia ringa-pa*
 this smell.bad-IPFV
 ‘This is stinking.’

Duna smell terms are thus unusual within the basic perception verbs in their polysemy as transitive and intransitive forms and in the inclusion of evaluative content in their meaning.

In natural discourse, they are probably more common as nouns (e.g., ‘a pleasant/nasty smell’) or as elements in complex constructions (e.g., verb serialization) than as simple final verbs, but this needs further investigation of corpus data.

5. Grammatical strategies

Having described the basic perception verb inventories of Guambiano and Duna, we now explore how grammatical structure participates in the encoding of perceptual experience. We first examine Viberg’s three semantic components—control, sense modality, and orientation—to determine whether and how they find expression at the grammatical level. Given that both Guambiano and Duna possess relatively parsimonious perception verb lexicons, with fewer lexical distinctions compared to languages such as English, morphosyntactic strategies may “take over” where lexically encoded contrasts end. We then look beyond Viberg’s framework to consider the grammatical expression of subjectivity in perception event descriptions.

5.1. Agency and control

Neither Guambiano nor Duna strongly lexicalize a controlled versus non-controlled distinction in their verb paradigm. This is not unusual in languages of the world—Norcliffe and Majid (2024a) report in their sample of 100 languages that for visual and auditory verbs, around 65% do not lexically distinguish along this dimension. Given that simple verb roots do not specify control, we can ask whether other constructional devices can serve this purpose instead. For Guambiano, and to a lesser extent Duna, the answer depends on the sense modality in question. For vision and audition, where agentive distinctions are not made lexically, context can do the work to convey these distinctions (although see also discussion of case-marking in Duna, below). This is evident in GPQ translations where linguistic contexts were designed to either convey active, controlled perception, or an unintentional experience. Agentive constructions were presented in “want to know” frames (e.g., *I saw a necklace in the grass. I wanted to know if it was valuable. I picked it up and looked at it*), while non-controlled perception events were presented in “suddenly” frames (e.g., *My friend and I were*

sitting by the river when **suddenly I saw a fish in the water**). In Guambiano, the same lexical verb *ash-* ‘see’ was used in both contexts:

(G18) *na kusru-a ash-ar*

1SG lift-SS see-EGO

‘I picked [the necklace] up and looked at it’

(G19) *na ash-ar kan piyau-wan*

1SG see-EGO one fish-ACC

‘I saw a fish’

Similarly, Duna *ke-* ‘see’ and *waki-* ‘hear, sense’ can be used to express both controlled attention and unplanned or undirected sensory experience, without further grammatical differentiation.

For taste and smell, by contrast, Duna and Guambiano have grammaticalized SERIAL VERB CONSTRUCTIONS to explicitly express agentive perception, and Guambiano also extends this pattern to touch. Serial verb constructions involve the sequence of two or more verbs within a single clause. These verbs act together to describe a single event or closely related actions. In the present context, comparing these kinds of constructions in the two languages reveals interesting language-specific conceptualizations of perception events and their subcomponents.

For Guambiano, in the case of touch and taste, the first verb in these constructions expresses what Pawley (2020) refers to as the “preparatory action” for the perception event and is inflected for the same subject marker (*-a*, indicating that the subject of the first and the second verbs is the same; see Floyd & Norcliffe, 2016); the second verb expresses the resultant perception (using the multisense perception verb *mirip* ‘to sense’). Thus, to actively feel by touch is ‘touch + sense’ (*kemira mirip*), and to actively taste is to ‘drink + sense’ (*mucha mirip*) or ‘eat + sense’ (*ma mirip*). These last two are notable in that they constitute a mini-paradigm of tasting types, based on the mode of ingestion.

The asymmetry between the senses exhibited here (the fact that active touch and taste have grammaticalized serial verb constructions, while vision and audition do not) possibly follows from differences in whether active perception is conceptualized as salient combinations of subevents, depending on modality. That is, intentional tasting typically involves the preparatory act of actively putting food or drink in the mouth, and active feeling requires exploratory touch, while active looking and listening do not have one obvious associated preparatory act.

In Guambiano, active smell also involves *mirip* ‘to sense’ as the second constituent of a complex predicate but differs structurally from active touch and taste in that the first verb *itri* ‘smell’ occurs as a bare root (it is not inflected for the same subject marker). When it occurs as a simple inflected verb, *itri-* has a phenomenon-oriented/sense emission meaning (e.g., the soup smells, see G46); it is not clear that this meaning is preserved in the serial verb construction, as this is unlikely to yield the agentive meaning of the construction as a whole. Possibly, the verb was originally ambi-valent (as English *smell* is), and the agentive construction has emerged by analogy with taste and touch.

In a similar (but not identical) vein to Guambiano, the Duna multisense verb *waki-* is also attested in serialization with the smell verb *kori-* where the subject exerts control over their action (D20).³

- (D20) *Ka kori waki-pa*
 2SG.ERG smell.good sense-IMP
 ‘You smell it!’

However, Duna situations involving active exploration or investigation more typically incorporate *ke-* ‘see’ as a final verb (D21–23), thus showing constructional similarity to Guambiano in the use of serial verb constructions but a departure in terms of verb choice. These serializations might also specify “preparatory actions” as discussed above, as in the expressions *neya ke-* (literally ‘eat see’) for “taste” and *suwa ke-* (‘hold see’) for ‘feel (at)’. In these situations, the visual semantics of *ke-* appear to be bleached or suppressed, and meanings to do with testing, trying, or investigating come more prominently to the surface. The lexical semantic relationship between *seeing* and *trying* has been widely documented across languages (Norcliffe & Majid, 2024b; Voinov, 2013), and in Papuan languages, vision verbs are frequently recruited in serial verb constructions to signal the conative modality (Foley, 1986, p. 152; San Roque & Schieffelin, 2019, pp. 356–357).

- (D21) *pisi hinia, nguni=rape peli=rape no kono neyata*
 fish this bad=UNC good=UNC 1SG know NEG-SEQ
kei=na, ka kori ke-pa
 be/stand.STAT=SPEC 2SG.ERG smell.good see-IMP
 ‘This fish, I don’t know if it’s bad or good, you smell it and see.’

- (D22) *Petrusi=ka wi=na hini=ta suwa ke=na*
 PSN=ERG possum=SPEC fur=OBL hold see=SPEC
 ‘Petrus is feeling at the fur of the possum.’

- (D23) *na supu=na neya ke-o, epo paye kone*
 1SG.ERG soup=SPEC eat see-PFV tasty good true
 ‘I tasted the soup, it was delicious.’

Agentive meaning in Duna can be reinforced through case-marking. Across languages, ERGATIVITY typically involves singling out the subject argument in transitive clauses for special treatment (e.g., unique case marking that cannot be applied to any other argument noun phrase (NP); special co-reference rules across clauses). In Duna, as with some other languages of New Guinea, ergative marking is pragmatically conditioned and associated with focus and increased semantic transitivity rather than being syntactically obligatory (San Roque, 2008; see also Donohue & Donohue, 1997). Thus, it is not the case that all subject NPs in transitive clauses must carry an ergative marker; rather, ergative marking can be applied to subjects to

emphasize agentive qualities (among other features).⁴ As hinted by (D21–23), ergative case marking may be especially appropriate in intentional, exploratory situations. For example, compare (D23), with a context of purposeful investigation (with ergative marking on the subject and serialization with *ke-*), to (D24), in the context of encountering hot chilli, with no intentional exploration, and no ergative marking.⁵

- (D24) *no lombo na=yanua, ko pi na=yarua=pe?*
 1SG chilli eat=SNS.IMPL 2SG also eat=SNS.C=Q
 ‘I ate (tasted) chilli, did you also eat (taste) chilli?’

A controlled or investigative meaning for basic perception verbs in Duna can thus be instantiated by several elements across the clause (serialization with another transitive verb, case-marking on the NP, and also individuation of the object as a definite NP). A special interpretation of the polysemous verb *waki-* ‘hear, sense’, as having the meaning ‘ask’, also seems to be supported through the use of ergative marking. Example (D25) shows *waki ke-*, ‘want to hear’, with no ergative marking present, whereas (D26–27), with ergative marking on the subject, were immediately interpreted by language consultants as having the meaning ‘ask’.⁶ In such examples, it seems the agency of purposeful listening is conceptually extended as the intent to elicit an aural stimulus (i.e., the spoken answer to a question) from another person.

- (D25) *No waki ke-nda*
 1SG hear see-INT
 ‘I want to try to **hear**.’

- (D26) *Na waki ke-nda*
 1SG.ERG hear see-INT
 ‘I will (try to) **ask**.’

- (D27) *Nane=na=ka waki=tia*
 boy=SPEC=ERG hear=PFV.VIS
 ‘The boy **asked**.’

Overall, these examples illustrate how morphological and constructional resources can contribute to the expression of control and intentionality in perception descriptions, and how this can in turn interact with the semantic potential of the verb.

5.2. Sense modality

*The eye is present for all five senses. There is an eye in the ear. There is an eye in the tongue. All parts must have an eye.*⁷ PK, from GPQ discussion.

Both Guambiano and Duna have MULTISENSE verbs. Across languages, these remain a puzzle as to whether they are best understood as polysemous (e.g., a verb with several related but distinct modality meanings) or as conveying a generalized meaning of perceptual experience (e.g., “sense, perceive, experience”). Regardless of the appropriate analysis, from a hearer’s perspective, the use of a multisense verb leaves modality information underspecified. In many instances, it is presumably communicatively unnecessary to be more specific, for example, if the focus is on the resultant state of perception (or knowledge obtained through perception), rather than the channel by which it was achieved. Nevertheless, different elements of the constructions in which such verbs occur often function to make modality information explicit. The nature of the stimulus object, for example, may be informative, where these are uniquely perceivable by a single modality. Thus, in Guambiano, a sentence like (G28), in which the stimulus is an auditory object, implies auditory perception:

(G28) *traseik-wan mir-ar*
 scream-ACC sense-EGO
 ‘I sensed a scream’

Similarly, in Duna, more specific sensory information can be supplied by nominal elements in the clause (D29).

(D29) *Julinda=kho ra-ta, mundu.hawu kori ndu waki-u*
 PSN=3SG sit/be-SEQ smoke smell.good one sense-PFV
ri=tia
 say=VIS.PFV.P
 ‘Julinda was there and she sensed a (pleasing) smoke smell she said.’

Guambiano exhibits an interesting restriction on the kinds of entities that can be selected as an object of a multisense verb, which has an impact on sensory interpretation. Many languages, including English, accommodate a wide range of direct objects in transitive perception verb constructions: One can, for example, perceive a perceptual signal (*see a light*, *hear a noise*, *smell an odor*), the source of a perceptual signal (*hear a bird*, *smell a skunk*), an entire event with spatio-temporal properties (*see the ball roll across the road*), or a more abstract proposition (*see that the floor needs sweeping*; see Viberg, 2001). In Guambiano, by contrast, what can be grammatically expressed as being directly heard, felt, or smelled is more limited (for gustation, the data are not clear). For these modalities, it appears that the concrete object producing the perceptual signal cannot stand in for the signal itself as a direct object of the verb. Thus, while an auditory object (e.g., a “shout” or a “noise”) can be directly heard (i.e., occur as a grammatical direct object), sources of sounds apparently cannot be—language consultant Luis Tunubalá (henceforth LT) considered simple sentences such as ‘I heard a bird’

(G30) infelicitous, instead offering (G31), in which the entire sound-making event is the direct object of the verb:

(G30) **na illi-wan mir-ar*
 1SG bird-ACC sense-EGO
 Intended: ‘I heard a bird’

(G31) *na illi reseik-wan mir-ar*
 1SG bird pray-ACC sense-EGO
 ‘I heard the bird praying’ (= singing)

Similarly, encoding ‘ants’ as a direct perceptual object was deemed infelicitous, in the translation of ‘I felt ants on my legs’ (G32–33); instead, a subordinate clause expressing the motion of the ants was offered. And for olfaction, it was not possible to simply ‘sense smoke’; instead, LT preferred a more complex construction involving a subordinate verb (i.e., one does not sense smoke, one senses when smoke is smelling):

(G32) **ayan-mera neisrip-yu mir-ar*
 ant-PL leg-LOC sense-EGO
 Intended: ‘I felt ants on my legs’

(G33) *ayan-mera neisrip-yu netrup-en mir-ar*
 ant-PL leg-LOC climb-SUB sense-EGO
 ‘I sensed when ants were climbing on my legs’

(G34) *Maruja neniwawa nakisr itrip-en mir-an*
 Maruja cooking smoke smell-NOM sense-NON.EGO
 ‘Maruja was cooking when she smelled smoke’
 [lit.: sensed when smoke was smelling, i.e., emitting a smell]

There are scattered reports of metonymy constraints in other languages, similar to those apparently exemplified by Guambiano. Pawley (2020) observes for Kalam (Madang, TNG) that “you cannot simply hear a sound...or an object that is the source of a sound. You can only hear something make a sound.” Expressing auditory perception therefore requires the use of a two-clause construction involving a verb of sound emission. Descriptions of olfactory perception in Kalam are similarly constrained to specify the perception of the odor (rather than simply its source). Tariana (Arawak) has a polysemous perception verb that can be used to express both sight and hearing, but, significantly, in its auditory use, it is grammatically constrained to select an auditory object (e.g., words, sounds, language, and a voice; Aikhenvald, p.c., cited in Wierzbicka, 2010, p. 80). In Avatime, the non-visual multisense perception verb *nu* can only select for direct objects that refer to an abstract

perceptual signal (nouns such as ‘noise’, ‘song’, ‘scent’, and ‘heat’). If a perceptual source is also mentioned in the clause, then it typically occurs as the possessor of the perceptual signal (e.g., ‘I sensed the dog’s scent’; van Putten, 2020).

Metonymy constraints reveal an asymmetry in the way the percept is treated in linguistic representations of perceptual events, depending on the sensory modality involved: While concrete objects can be directly “seen,” they cannot be directly heard or smelled, and so forth. These constraints may have a communicative basis: It is notable that for the languages for which such restrictions or preferences have so far been reported, they are associated with multisense perception verbs, so that metonymy constraints may help to clarify an intended sensory interpretation where ambiguity is present (cf., Norcliffe & Majid, 2024a). Potentially, the verbal inventory and metonymic constraints could work together to mutually re-enforce the broad distinction between “vision” and “else” that surfaces across linguistic elements. Metonymic constraints may additionally tap into a particular conceptualization of what “counts” as an object of perception, grounded in the phenomenological properties of the different senses. Whereas visual experience readily supports object individuation and spatial anchoring, non-visual modalities often foreground processes or events (e.g., a sound being made, a smell being emitted). These differences may be linked to the temporal unfolding and diffuseness of non-visual perceptual stimuli, leading to constraints on object selection in clause structure. Ascertaining the typological extent of such constraints, the factors motivating them, and how they interact with the perception verb inventory are important areas for future research

Metonymy constraints do not appear to be so strongly present in Duna, but there is potentially a preference for multi-clausal structures in describing perception, consonant with the Guambiano constructions in (G33–34). For many of the GPQ responses, PK chose to present the perception event not as a simple clause (e.g., two noun phrases and a final verb) but as a constellation of subevents. These included, for example, a subordinated clause describing an aural predicate (D35), and an introductory (non-finite) perception verb that provides a frame for the perceived clause (D36). For this latter construction, the percept is not clearly integrated as a grammatical object. Rather, the placement of the verb *waki-* ‘sense’ sets the scene for Julinda’s perceptual experience, and the evidential at the end of the clause (see Section 5.3) cements the implication.

(D35) *Rosa=kho heka haka=yatia-ko waki-u (ri=tia)*
 PSN=3SG bird speech=IPFV.SNS.P-SUB sense-PFV say=PFV.VIS.P
 ‘Rosa heard the birds calling (she said)’

(D36) *Julinda=kho waki-yaroko, ndune ndu sia=ta kerema=yatia*
 PSN=3SG sense-SW.SIM thing one leg=OBL ascend=IPFV.SNS.P
 ‘While Julinda was sensing, something was climbing up her legs (sensed).’

Finally, the GPQ yielded an intriguing example of how spatial descriptors can support the modality interpretation of multisense verbs in Guambiano. In a context where somebody has

kicked a ball at the speaker who had their back turned, the sentence ‘I didn’t hear it but I felt it’ was translated into Guambiano in the following way:

- (G37) *eshelu chu kutri mir-m-ur inchenti sil pu-in-ti mir-ar*
 ball where from sense-NEG-EGO but correct arrive-RES-RESTR sense-EGO
 ‘I didn’t sense from where, but I just sensed it arrive’

In this construction, the modality contrast is expressed via spatial deictic distinctions—I did not sense from where, but I sensed it arrive (i.e., on my head). This contrast captures the salient difference between hearing as a distal sense and touch as a proximal sense: ‘from where’ indicates a location away from the speaker’s body, while the verb *pu* ‘arrive’ indicates proximity to/contact with the speaker. While we see this conceptual distinction emerging in an ad hoc constructional choice in Guambiano, in some languages the distal/proximal distinction is grammaticalized at the lexical level. For example, the language Wichí Lhamtés Vejoz (Matacoan) has a perception verb root that in combination with a proximal locative marker means ‘to feel (tactile)’ and with a distal locative marker means ‘to hear’ (Nercesian, 2014, p. 238). The spatial encoding of perception has been addressed in only a handful of studies to date (e.g., Huumo, 2010; Kawachi, 2020; Wnuk, 2022, for spoken languages; Takashima, 2019; Zeshan & Palfreyman, 2019, for sign languages; see Majid & Norcliffe, 2026, for discussion) and is in need of systematic typological study.

The examples discussed so far illustrate how constructional or other semantic resources can narrow in on the perceptual modality of multisense verbs. We also find contrastive cases where unimodal perception verbs can be implicated in other sensory experiences through constructional context. In Duna, the verb *ke-* ‘see’ is sometimes present in descriptions of non-visual experiences, for example, in (D38), which describes auditory investigation (see also Section 5.1). The thing being investigated is specified as a non-visual item through the use of the sensory evidential = *yanua* (see Section 5.3.1), reinforced by the sensory verb *waki-* ‘hear, sense’ in the next clause.

- (D38) *ho=yanua aki=pi ke-nda, waki peli-pa*
 ho=SNS.IMPL what=OPIN see-INT hear well-IMP
 ‘To see (find out) what this is, listen carefully.’

In such cases, *ke-* is likely conveying meanings that are not vision-specific, but more to do with active investigation, cognition, or awareness, as already discussed. Modality mismatches in these kinds of examples may even reinforce or facilitate polysemous meanings over time (San Roque, Kendrick, Norcliffe, & Majid, 2018).

5.3. Orientation to the stimulus

So far, we have looked at perception event descriptions that use perceiver-oriented verbs and thus foreground the perceiver as a conceptually and grammatically essential participant: the subject argument of the clause. An alternative approach in talking about perception is to focus on the stimulus and its sensory properties. In English, as we have seen, this can

be accomplished by the use of stimulus-oriented perception verbs, such as *look* in *the moon looks bright* and *sound* as in *the singer sounds hoarse*. In these examples, the stimulus is the grammatical subject, supporting the elaboration of perceptual detail, while the perceiver is left unexpressed (although typically associated with the speaker).

As seen in Section 4, Guambiano and Duna have only a limited range of stimulus-oriented basic perception verbs; in this corner of the lexicon specifically, the capacity to express the stimulus as the subject is virtually unsupported. Particular sensory properties of a stimulus subject might be expressed using other lexical resources, such as semantically specific verbs of appearance, noise-making, and smell. For Guambiano and Duna, grammatical resources offer further alternatives to reference basic perception without making the perceiver explicit through the use of EVIDENTIAL MARKERS.

Evidentials are commonly defined as grammatical elements that express the information source for a proposition (Aikhenvald, 2004; Willett, 1988), for example, they may indicate if the activity described by a clause was directly witnessed (so-called “direct” evidentials) as opposed to learned through hearsay. Debate is ongoing as to the best way to understand evidential categories and their functions (e.g., Anderson, 1986; Bergqvist & Grzech, 2023; Boye, 2012; Lazard, 1999; Murray, 2010; Mushin, 2001). Direct evidentials are especially relevant in the study of perception grammar, as they can encode or imply the sensory channel through which information was obtained.⁸ For example, many languages with evidential systems include grammatical markers for visual apprehension (e.g., seeing a fire), contrasting with a marker that indicates other sensory experience (e.g., hearing or smelling a fire). Evidentials, then, can be viewed as an alternative or supplementary way to convey perceptual experience, in addition to verbs or other lexical items.

Guambiano has two direct sensory evidentials. The morpheme *-tre* indicates that the speaker has visual evidence for their assertion (G39) and *-shi* indicates non-visual, sensory experience (G40) (see Norcliffe, 2018; Vásquez de Ruíz, 2007).

(G39) *chucha amtru-tre!*

possum come-VIS

‘A possum is coming’ (seen)

(G40) *chucha amtru-shi!*

possum come-NONVIS

‘A possum is coming’ (sensed; e.g., by hearing its movements, or smelling it)

Evidentials can serve a similar function to basic stimulus-oriented verbs in languages such as English, as they present the percept as the subject of a clause (and, concomitantly, suppress or demote the perceiver). In Guambiano questionnaire responses, evidentials were common in translations for scenarios based on monovalent stimulus-oriented English perception verbs such as *look* or *feel* (or equivalent Spanish constructions). In (G41), for example, the non-visual sensory evidential *-shi* on the verb ‘be’ indicates that the water appears clean based on non-visual properties (e.g., touch, taste, or smell); such a construction would be

appropriate where English, for example, might rely on a stimulus-oriented verb (the water felt/smelled/tasted clean).

- (G41) *pi-pe chishchik ki-shi*
 water-TOP clean be-NONVIS
 ‘The water was clean’ (sensed; e.g. isn’t greasy; by touching, tasting, smelling)

Example (D42) shows the Duna visual evidential, and (D43–45) illustrate non-visual forms. The Duna GPQ responses include both verbal predicates (e.g., ‘be’) and non-verbal predicates (e.g., adjectives such as *paye* ‘good’) marked with evidentials; virtually any property can thus be highlighted as a visual or non-visual sensory experience, despite the fact that Duna does not have basic stimulus-oriented perception verbs to serve this function.

- (D42) *no hi=ta iwa-yaroko, eke ombene ka=rua.*
 1SG outside=OBL go.out-SW.SIM moon full be/stand=STAT.VIS.P
ke-yaroko paye kone=rua
 see-SW.SIM good true=STAT.VIS.P
 ‘As I was going outside, the moon was full (seen). As I was looking, it was very beautiful (seen).’

- (D43) *Ah, lombo ra=yanua*
 Excl chilli be/put=SNS.IMPL
 ‘Oh, there’s chilli in this (sensed).’

- (D44) *no na-yeroko, epo kone=yatia*
 1SG eat-SW.SIM tasty true=IPFV.SNS.P
 ‘I was eating, [it was] delicious (sensed).’

- (D45) *Ho-ne, se=ra=ka nga-yaroko,*
 ho-? area=SHRD=PLC go-SW.SIM
ringa nguni kone=yatia ri=tia.
 smell.bad bad true=SNS.IPFV.P say=VIS.PFV.P
 ‘As [he] approached the area, the stink was terrible (sensed) [he] said.’

Finally, we note that evidentials in Guambiano not only provide a strategy for perception descriptions that are oriented to the stimulus but also give an additional facility to tap into the multi-sensory nature of experience (Calvert, Spence, & Stein, 2004). In Guambiano, the non-visual sensory evidential can combine with phenomenon-oriented perception verbs (G46). In this example, the sensory meaning of the evidential and the predicate are aligned—the

olfactory meaning of the predicate is compatible with the non-visual sensory meaning of the evidential:

- (G46) *mamik trinpikalik itri-shi*
 soup bad smell-NONVIS
 ‘The soup smelled bad’

More intriguingly, Guambiano grammar perhaps gives the option to convey two modalities at once. In (G47) and (G48), the visual evidential on the finite auxiliary verb *mismatches* the nominalized smell verb. According to LT, the use of the visual evidential in such a construction would only be possible if the speaker is looking at the percept (while also perceiving its fragrance). If someone comes into the room holding the flower or the bread, which the speaker smells, but his or her back is turned, then the visual evidential would not be acceptable:

- (G47) *u-pe kuichik itr-ap-ik ki-tre*
 flower-TOP sweet smell-DUR-NOM be-VIS
 ‘The (visible) flower is sweet smelling’

- (G48) *pitri-pe maichik itr-ap-ik ki-tre*
 bread-TOP delicious smell-DUR-NOM be-VIS
 ‘The (visible) bread is delicious smelling’

Similarly, in (G49), the use of the visual evidential on the auditory verb ‘to sound’ is only possible if the speaker sees that the radio is turned on. If the speaker only hears the radio, then the visual evidential would not be felicitous:

- (G49) *wam-wit-ap-ik wantri-tre*
 word-throw-DUR-NOM sound-VIS
 ‘The (visible) radio sounded.’

Such constructions are typologically unusual and appear to convey multisensory experiences with modality information distributed across different parts of the construction. To date, they have only been elicited in Guambiano in the context of the GPQ; more work is needed to ascertain how typical these kinds of multi-modal descriptions are in natural speech.

6. Grammatical sensitivity to subjectivity in perception event descriptions

Our study of Guambiano and Duna perception grammar has focused on the three elements of perception events highlighted in the Vibergian framework: control, modality, and orientation. Within this framework, we have recognized that a perceiver may have gradable agency but have otherwise treated this as a basically undifferentiated role: the person who sees, hears, feels, and so forth. In this last section, we move beyond the original framework to highlight how the perceiver role not only interfaces with the semantics of the event and its syntactic presentation (e.g., perceiver as transitive subject) but must also be integrated into the speech situation.

Perception is ultimately a subjective experience: Only the wearer knows where the shoe pinches. Or as Duna consultant PK elaborated in discussion of the GPQ, “another’s eyes are not your eyes, and you can’t combine them to look at things. They are different eyes.” Thus, when a speaker talks about their own subjective sensations (e.g., *I taste chilli in the soup*), they are in a position of privileged access: They can assert this proposition with more authority than when stating another person’s experiences. In questions, however, privileged access is usually assigned to the addressee (e.g., *Did you taste chilli in the soup?*). The *epistemic authority* (Hargreaves, 2005) in the speech situation—the person who the speaker presupposes or claims to have privileged knowledge of the experience under discussion—thus often aligns with the first person in statements and the second person in questions. Language may reflect such asymmetries of access, showing “person sensitive” distributions of lexical and grammatical categories (Post, 2013; San Roque, Floyd, & Norcliffe, 2018). In Japanese, for example, “private” predicates of emotion and sensation are restricted to first person statements and second person questions, requiring additional constructional support when used in the third person (see e.g., Hirose, 2000; Majid, 2012a).

The typologically rare category of *egophoricity* is a person-sensitive grammatical category *par excellence*. Egophoric morphology tracks the epistemic authority in the speech situation, aligning with first-person-oriented statements and second-person-oriented questions (San Roque, Floyd, & Norcliffe, 2018). Both Duna and Guambiano, in fact, show deep integration of egophoricity in their grammar such that speakers are routinely conveying assessments of epistemic authority through verbal inflection (Norcliffe, 2018; San Roque, 2018). For Guambiano, this is a semantically general habit; for example, all first-person verbs take the egophoric marker *-ar* in statements regardless of the nature of the event. For Duna, however, some egophoric morphology is sensitive to the experiential nature of the predicate. Thus, discussion around the GPQ suggested that the verbs *ke-* ‘see’ and *waki-* ‘hear/sense’ are part of a very small class of verbs that can be marked with a special egophoric present inflection, *-i*. Example (D50) shows that this inflection suitable for use in situations where the perceiver is also the epistemic authority. Contrastively, clauses in (D51) illustrate consultants’ assessments that the perception verb *waki-* cannot be combined with present *-i* in other circumstances.

- (D50 a) *no* *waki-i*
 1SG hear-PRES
 ‘I’m listening.’
- b) *ko* *waki-i=pe?*
 2SG hear-PRES=Q
 ‘Are you listening?’
- (D51 a) **ko* *waki-i*
 2SG hear-PRES
 (you are listening)
- b) **no* *waki-i=pe*
 1SG hear-PRES=Q
 (Am I listening?)

- c) **Julinda =kho waki-i*
 PSN=3SG hear-PRES
 (Julinda is listening)

Both *ke-* and *waki-* also participate in a special construction that is used for asking about the perceptual (or cognitive) judgment of the addressee, usually encouraging them to share a view of the speaker (similar to appeals in English such as *d'ya see?* or *don't you reckon?*); see also San Roque et al., 2012). This potentially reflects the special relationship that references to perception can have in establishing (or disputing) common ground between speaker and addressee and the frequent extension of perception verbs to discourse markers in the world's languages (e.g., Kendrick, 2019; San Roque, Kendrick, Norcliffe & Majid, 2018; Sidnell, 2007).

Responses from the GPQ also illustrate the flipside of egophoric sensitivity: Talking about someone *else's* perception may require additional mediation to sound natural. In Duna, in many of the third-person examples offered for translation, PK spontaneously added the phrase *ritia* '[they] said' to the end of the response, as in (D52). Framing the perception event as a speech report from the perceiver makes it more pragmatically acceptable (cf., e.g., Schieffelin, 2008).

- (D52) *Julinda =kho ipa.wena ndu ipa=ta ngo-rane ke-o ri=tia*
 PSN=3SG fish one water=OBL go-SW.SEQ see-PFV say=VIS.PFV
 'Julinda saw a fish swimming in the water, she said.'⁹

In Guambiano, simple statements of third-person perception were more readily produced in translations. Interestingly, however, questioning the addressee about the perception of a third person required further constructional support. For these, LT considered sentences less acceptable when the perception verb was in the simple interrogative form, instead preferring constructions with a final interrogative-inflected auxiliary *chintera*, which produces a dubitative reading (G53–54). This suggests a disinclination to attribute to the addressee direct knowledge of a third person's perceptual experiences:

- (G53) *iik-pe piyau-wan asa chinter-a'?*
 DIST.DEM-TOP fish-ACC see do-INT
 'Could it be that he/she saw the fish?'
 (G54) *iik-pe traseik-wan mira chinter-a'?*
 DIST.DEM-TOP scream-ACC sense do-INT
 'Could it be that he/she heard the scream?'

The Duna and Guambiano examples illustrate two ways that grammatical phenomena interface with the subjective nature of perception. On the one hand, we see overt marking that reflects the unique access a speaker has on their own sensory experience (and that a ques-

tioned addressee can be assumed to have, cf. San Roque, Floyd, & Norcliffe, 2017). On the other, we see how making the leap to assert or question another person's perception can require extra structural work, such as the introduction of a reported speech frame or dubitative construction.

7. Conclusion

The comparative study of perception event encoding in Guambiano and Duna reveals diverse morphosyntactic resources that are drafted into expressing perceptual experience. These resources form part of the mosaic of linguistic diversity, highlighting the risk of relying on a single language to speak for all of us (Blasi, Henrich, Adamou, Kemmerer, & Majid, 2022; Ghai et al., 2025; Majid, 2023).

Viberg's three core semantic components of control, sense modality, and orientation are given fairly short shrift in the lexical inventories of both Duna and Guambiano: Control is not contrasted in simple lexemes at all, certain sense modalities are collapsed into multisense verbs, and stimulus-oriented verbs are found for just one or two senses. Other elements, such as verb serialization and case-marking, support contrastive expressions and participate in polysemous extensions of basic perception verbs. The spatial encoding of sensory distinctions, glimpsed in Guambiano's use of spatial deictic contrasts to distinguish auditory from tactile perception, is a promising avenue for cross-linguistic exploration.

The grammatical encoding of perception in these languages suggests asymmetries across the sense modalities, possibly grounded in the phenomenological properties of different senses (Strik Lievers & Winter, 2017) and pointing to questions for future systematic research. Serial verb constructions in both languages express controlled perception only for touch, taste, and smell but not for vision and audition. This possibly reflects an event structural asymmetry between the senses—while taste and touch involve salient preparatory actions (e.g., ingestion, contact), controlled vision and audition lack obvious preliminary actions. Guambiano's metonymy constraints further reveal that while concrete objects can be directly “seen,” they cannot be directly “heard” or “smelled.” This pattern is also found in other languages (Pawley, 2020; van Putten, 2020) and plausibly reflects communicative motivations to do with disambiguation (cf., Norcliffe & Majid, 2024b) alongside perceptual differences between different kinds of stimuli—while visual stimuli are frequently discrete and spatially bounded, auditory and olfactory stimuli unfold temporally and spatially and may therefore be most likely to be grammatically encoded as events. If these asymmetries do indeed tap into phenomenological differences between the senses, we should expect to observe further parallelisms across unrelated languages.

The relationship between evidential systems and perception grammar also deserves systematic investigation. For both Guambiano and Duna, evidentials emerge as expressive tools for sensory description, potentially sharing functions with stimulus-oriented verbs; this requires exploration in contextualized discourse. The finding that Guambiano can utilize evidentials in combination with perception verbs to express multiple sense modalities at once (e.g., visual evidentials with olfaction predicates) is intriguing, potentially showcasing one way that gram-

mar can reflect the inherently multisensory nature of embodied experience. This is an exciting and understudied area in sensory linguistics.

Beyond Viberg's original framework, the data highlight the fundamental role of subjectivity in perception grammar. The speakers who shared their time and knowledge to undertake the questionnaire showed a sensitivity to the epistemic asymmetries inherent in some of the tasks, illustrating, in their use of egophoric and evidential strategies, language-specific grammatical solutions to the universal problem of how to talk about perception in a way that both preserves and bridges individual experience.

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Notes

- 1 The Totoró variety, now spoken by only a handful of speakers, has been described in most detail (Gonzales Casataño, 2019); it is sometimes considered a separate language from Guambiano (Guambía Namtrik), and the two varieties have separate glottocodes (toto1306 and guam1248).
- 2 The language of each example sentence is indicated as: E = English; D = Duna; G = Guambiano. For Guambiano and Duna interlinear grammatical glosses, abbreviations generally follow the Leipzig Glossing Rules (<https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>), with additions: EGO, egophoric; OBL, oblique; PSN, personal name; SEQ, sequential; SHRD, shared knowledge; SNS, sensory; SW.SIM, switch simultaneous; UNC, uncertain; VIS, visual.
- 3 It seems likely this is preferred to simple imperatives with smell verbs, which our Duna consultant accepted as potentially sayable but did not enthusiastically endorse. Overall, then, the Duna smell verbs can potentially be understood as lower in inherent agentivity or control than the clearly bivalent “hear/sense” verb, as is also suggested by the ambivalent nature of these verb roots.
- 4 In Duna, “ergative” marking can also be applied to intransitive subjects (S), departing quite a distance from the traditional definition of ergativity, which is restricted to transi-

- tive subjects (A) only. However, Duna does fit broadly within the descriptive tradition of “optional ergativity” in the New Guinea region (e.g., Riesberg, 2018)
- 5 The sensory evidentials used in this example also contribute to the meaning of involuntary sensation; see San Roque (2008)
 - 6 The visual evidential in (D27) may also support the interpretation as ‘ask’ here, as asking is potentially a more visually accessible activity than hearing. See Section 5.3.
 - 7 Tok Pisin original: *Ai bai stap lo olgeta five senses. Ai stap lo yau. Ai stap lo tang. Olgeta hap em mas i gat ai ya.*
 - 8 Indirect evidentials may also be highly relevant to perception in a number of ways. For example, from a comparative perspective, complement constructions such as *I saw that ...* or *I heard that...* in English may express inferential and reported meanings, respectively, that is, sharing functions with inferential/reported morphemes in languages with grammaticalized evidentiality. This is beyond the scope of the present paper.

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